

Lilly Ludlow

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Objective

Highly motivated industrial engineering student with a minor in environmental and ecological engineering and electrical and computer engineering, seeking a summer internship to leverage academic background and hands-on experience to contribute to innovative projects, while gaining relevant industry experience.

Proficient in: R, C, C#, Python, MATLAB, JavaScript, Verilog, SystemVerilog, LaTeX, VBA, HTML, SQL, PHP, .NET, LabVIEW, LTSpice, KiCad, Excel, AutoCAD, GitHub, Embedded System Design, API Creation and Design, Data Structure Manipulation, Machine Learning

Education

Purdue University – West Lafayette, IN	Expected Graduation: December 2026
- Major: Industrial Engineering - Minors: - Environmental and Ecological Engineering - Electrical and Computer Engineering - GPA: 3.31	

Engineering Experience

Kohl’s Corporate – Supply Chain Industrial Engineering Intern	May 2026 – August 2026
- Analyzed Kohl’s non-sort operations leveraging key performance metrics to identify opportunities for improvement - Led biweekly meetings with 3 external vendors to evaluate and develop solutions tailored to identified areas of improvement - Estimated 4 labor models to determine the performance of each technology and conduct a cost-benefit analysis to quantify ROI, determine solution feasibility, and make several relevant recommendations for the process	
Hormel Foods Progressive Processing – Industrial Engineering Intern	May 2025 – August 2025
- Improved ergonomic efficiency through the elimination of 3 ergonomic risk factors by redesigning production processes and creating 15 supporting educational materials - Conducted 12 time studies to refine 7 existing work standards, improving efficiency and job loads in 3 departments - Collaborated with cross-functional teams to determine optimal machine settings through center lining, enhancing process stability and reducing variability and saved \$302,000 annually	
BorgWarner Technical Center – Test Development Engineering Intern	May 2024 – August 2024
- Designed an advanced control algorithm in C#, integrating diverse control techniques that reduced time for test completion by 87%, which significantly enhanced testing efficiency and accelerated project timelines - Analyzed historical test data trends to design 5 complex adaptive mathematical models regarding variable interactions to develop a precise data model and significantly enhance the control algorithm’s accuracy and reliability - Collaborated with a mentor to validate algorithm performance through comparison of the original algorithm through various testing methods with the proposed algorithm to determine algorithmic improvements in functionality	
Marion Utilities Office – Engineering Intern	May 2023 – August 2023
- Implemented multiple data analysis algorithms in various software platforms to maximize plant efficiency - Developed an energy map of 33 pump and motor systems to increase sustainable power output via an anaerobic digester - Collected and analyzed waste sample settling velocities at several points in treatment to design a program that generates an ideal plant pumping rate using Excel as a user interface and completing the project in VBA	

Career History

Purdue University Residences – Residence Education Assistant	May 2025 – Now
- Supervised and supported a team of 12 Resident Assistants through one-on-one biweekly meetings, monthly staff training, semesterly performance evaluations, and leadership in weekly staff meetings - Served as primary on-call crisis responder across four residential buildings, managing student concerns, policy enforcement, and follow-up while ensuring a safe and inclusive residential community for over 1,000 students and 50 RA’s - Collaborated with professional staff to lead and implement community-building initiatives focused on student development	
Purdue University Residences – Residential Assistant	January 2024 – May 2025
- Cultivated relationships with 50 first-year female engineering students to promote an inclusive, uplifting environment - Managed various administrative responsibilities, such as resident mediations, report writing, budget-making, event planning, and participating in an on-call rotation to protect resident health, safety, and well-being - Facilitated discussions with first-year students to address academic challenges and ensure support via campus resources	
Electrical Engineering Fundamentals 1 Lab - Teaching Assistant	January 2024 – May 2024
- Instructed students in constructing complex circuits for the first time, highlighting important electrical fundamentals - Reviewed and analyzed the lab reports of 80 students each week to assess their understanding of key concepts - Examined 40 circuits each week to identify issues, debug results, and address areas where students lacked comprehension	